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CYBERSECURITY VIRTUAL INTERNSHIP PROGRAM 2025

REPORT OF PROBLEM STATEMENT FOR CYBER SECURITY STREAM

Cyber Shield : Defending the network

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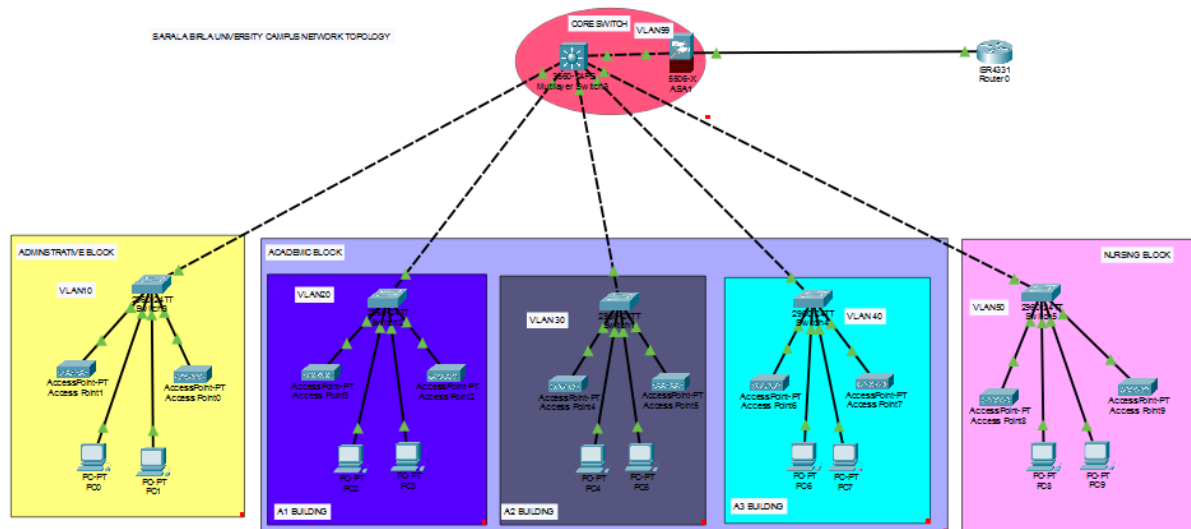
Cybersecurity Industry Project - Part 1: Cyber Shield: Defending the Network

1. Introduction

This report details the findings and recommendations from an internal red team assessment of the college campus network infrastructure. As part of the Cisco NetAcad Cybersecurity course, the objective was to analyze the existing network, identify security controls, assess their effectiveness against current threats, and pinpoint potential weaknesses. The assessment aimed to provide real-world insights, considering constraints such as tight budgets, limited staff, and security often being an afterthought.

2. Network Topology Diagram

The following diagram, created using Cisco Packet Tracer, provides a visual representation of the college campus network infrastructure, including devices, zones, and identified attack surface areas.



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Diagram Key Components:

- **Core Switch (CORE_SWITCH):** A Cisco 3560 Multilayer Switch, acting as the central point for inter-VLAN routing and connecting to all building access switches and the ASA Firewall.
- **ASA 5505 Firewall:** Positioned between the Core Switch and Router0, it controls traffic flow and performs Network Address Translation (NAT) for Internet access.
- **Router0 (ISP Simulation):** Represents the connection to the Internet Service Provider (ISP).
- **Access Switches (SW_ADMIN, SW_A1, SW_A2, SW_A3, SW_NURSING):** Dedicated switches for each building (Admin Block, A1 Building, A2 Building, A3 Building, Nursing Block), connecting end-user devices.
- **End Devices (PCs, Access Points):** Represent student and faculty workstations and wireless access points within each building.
- **VLANs (Trust Zones):**
 - **VLAN 10 (Admin Block):** 192.168.10.0/24 subnet.
 - **VLAN 20 (A1 Building):** 192.168.20.0/24 subnet.
 - **VLAN 30 (A2 Building):** 192.168.30.0/24 subnet.
 - **VLAN 40 (A3 Building):** 192.168.40.0/24 subnet.
 - **VLAN 50 (Nursing Block):** 192.168.50.0/24 subnet.
 - **VLAN 99 (ASA Transit):** 192.168.100.0/30 subnet, serving as the transit network between the Core Switch and the ASA Firewall.

3. Security Assessment Report

This section highlights identified security risks, provides a comprehensive attack surface analysis, and suggests solutions and countermeasures to mitigate these risks.

3.1 Network Segmentation and Trust Zones

The network utilizes VLANs to segment different functional areas (buildings). Each building's subnet is assigned to a specific VLAN, and inter-VLAN routing is performed on the Core Switch.

- **Trust Zones Identified:**

- **Internal Campus Networks (VLANs 10, 20, 30, 40, 50):** These represent the primary operational zones for administrative staff, various academic buildings, and the nursing block. All devices within these VLANs are considered to be within a higher trust zone relative to the internet.
- **DMZ/Transit Zone (VLAN 99):** This small, dedicated subnet serves as a transit network between the Core Switch (internal network) and the ASA Firewall, effectively acting as a demilitarized zone for the firewall's internal interface.
- **External/Untrusted Zone (Outside ASA to Router0/ISP):** This represents the Internet connection, which is considered completely untrusted.

3.2 Existing Security Controls

The following security controls have been identified in the current network infrastructure:

- **VLANs for Segmentation:** Network traffic is segmented by VLANs, separating broadcast domains and logically isolating different departments/buildings. This helps in containing network issues and can limit the scope of an attack.
- **Multilayer Core Switch (Inter-VLAN Routing):** The Core Switch is configured for interVLAN routing, allowing communication between different VLANs. While this enables necessary communication, the lack of granular access control lists (ACLs) directly on the SVIs could be a weakness.
- **ASA 5505 Firewall:** The ASA acts as a perimeter firewall, separating the internal network from the untrusted external network (Internet). It is configured with:
 - **Security Levels:** inside interface (GigabitEthernet0/1) with a security-level of 100 (trusted), and outside interface (GigabitEthernet0/0) with a security-level of 0 (untrusted). This implicitly allows traffic initiated from higher security levels to lower security levels.
 - **Dynamic NAT:** Translates private IP addresses of internal networks to the public IP address of the ASA's outside interface, enabling Internet access while hiding internal network topology.
- **DHCP Services on Core Switch:** The Core Switch provides DHCP services to all VLANs, automatically assigning IP addresses to devices. While convenient, the configuration does not explicitly include security features like DHCP snooping on access switches, which could prevent rogue DHCP servers.

3.3 Attack Surface Analysis

Based on the network topology and current security controls, the following potential weaknesses and attack surface areas have been identified:

- **Flat Zones (Lateral Movement):**
 - **Within VLANs:** While VLANs segment the network, devices within the same VLAN can communicate freely. If one device in a VLAN is compromised, an attacker can easily move laterally to other devices within that same VLAN.
 - **Inter-VLAN Communication:** The Core Switch facilitates inter-VLAN routing without explicit ACLs to restrict communication between specific VLANs. This means that a compromised device in one VLAN (e.g., A1 Building) could potentially access resources in another VLAN (e.g., Admin Block) if not further restricted. This is a significant "flat zone" concern that allows lateral movement across different logical segments.
- **Firewall Policy Granularity:**
 - The ASA firewall primarily performs NAT and relies on security levels. While it prevents direct unsolicited inbound connections from the Internet, the absence of specific access control policies (ACLs) on the ASA to control *internal* traffic between VLANs (if any were to be routed through it) or more fine-grained outbound filtering means that internal threats could still propagate or bypass less restrictive controls.
 - The ASA's default behavior allows return traffic for connections initiated from a higher security level to a lower security level. While generally desired, it means all outbound connections are implicitly allowed.
- **Lack of Intrusion Detection/Prevention Systems (IDS/IPS):** There is no mention of an IDS/IPS deployed. This means that the network lacks the ability to detect and prevent

malicious activities such as malware propagation, unauthorized scanning, or attack signatures in real-time.

- **Absence of Authentication Servers/802.1X:** No authentication servers (like RADIUS or TACACS+) or 802.1X port-based authentication are mentioned. This implies that any device connecting to an access switch port would gain network access without strong authentication, making it easy for unauthorized devices to join the network.
- **Wireless Security:** While Access Points are present, the specific security configurations (e.g., WPA3, enterprise authentication) are not detailed. Open or weakly secured Wi-Fi networks could be a significant entry point for attackers.
- **Endpoint Security:** The scope of this part doesn't cover endpoint security, but it's a critical component. Without endpoint protection (antivirus, host-based firewalls, patch management), individual PCs remain vulnerable even if network perimeter security is strong.
- **Lack of Logging and Monitoring:** The current configuration does not explicitly include detailed logging mechanisms beyond default device logging, nor is there a centralized monitoring solution (e.g., SIEM). This makes it difficult to detect, investigate, and respond to security incidents.

3.4 Attacker Target Prioritization and Countermeasures

Given the identified weaknesses, an attacker would likely target the following first, and here's how to stop them (assuming tight budgets, limited staff, and security as an afterthought):

1. Target: Lateral Movement within and between VLANs (Flat Zones).

- **Attacker Goal:** Once inside any VLAN (e.g., via a compromised student laptop or a social engineering attack), the attacker would exploit the flat network to move to more sensitive VLANs like the Admin Block or target other devices within the same VLAN.
- **Countermeasures (Risk-Based & Practical):**
 - **Implement VLAN ACLs on Core Switch SVIs:** This is the most critical and budget-friendly immediate countermeasure. Implement granular Access Control Lists (ACLs) on the Core Switch's VLAN interfaces (SVIs) to restrict specific traffic flows between VLANs. For example, explicitly deny or permit only necessary services (e.g., allowing Admin VLAN to access specific server ports only, blocking inter-student VLAN communication entirely).
 - *Policy Example:* Deny traffic from student VLANs (A1, A2, A3, Nursing) to the Admin VLAN (10) unless absolutely necessary for specific services.
 - *Justification:* Directly addresses lateral movement, limits attack blast radius, uses existing hardware.
 - **Port Security on Access Switches:** Configure port security on all access switch ports connected to end devices. This will limit the number of MAC addresses allowed per port and can shut down ports if violations occur, preventing unauthorized devices from connecting.
 - *Justification:* Low cost, effective against rogue devices and MAC spoofing.
 - **Implement ACLs on Access Switch Uplinks (if practical):** While trunk links, if a specific VLAN should *never* egress from an access switch (e.g., if a new untrusted VLAN were introduced), an ACL could be applied on the trunk. This

is less practical for existing segmented VLANs but an option for future expansion.

2. **Target: Lack of Basic Network Access Control.**

- **Attacker Goal:** Connect an unauthorized device (e.g., a personal laptop with malicious tools) to an open network port and gain immediate network access.
- **Countermeasures (Risk-Based & Practical):**
 - **Stronger Physical Security:** Ensure network closets and active equipment areas are secured with locks. This is a fundamental but often overlooked physical security control.
 - **Consider Basic 802.1X (Future/Phased Approach):** While potentially more complex, even a basic 802.1X deployment (e.g., using local authentication on switches or a simple RADIUS server) would significantly enhance network access control. Start with critical areas like the Admin Block.
 - *Justification:* Prevents unauthorized devices from connecting, even if port security is bypassed. Requires more planning but offers high security.

3. **Target: Perimeter Insecurity (Lack of deeper inspection).**

- **Attacker Goal:** Exploit vulnerabilities in legitimate outbound traffic or bypass basic security levels of the ASA.
- **Countermeasures (Risk-Based & Practical):**
 - **Implement Basic ACLs on ASA (if not using specific security policies):** While security levels implicitly handle some traffic, explicitly defining permitted outbound services (e.g., only HTTP/S, DNS) on the ASA would enhance security and prevent unknown protocols from traversing the firewall. This adds a layer of egress filtering.
 - *Justification:* Improves control over outbound traffic, helps prevent malware callbacks.
 - **Consider Stateful Packet Inspection (already present to some extent with ASA):** Leverage the ASA's stateful inspection capabilities to ensure only legitimate return traffic is allowed. (This is generally inherent to the ASA's operation).
 - **Introduce DNS-based Filtering (for web content):** This is a low-cost, effective way to block access to known malicious domains and potentially unwanted content categories at the DNS level. This can be implemented either on the internal DNS server (if one exists) or by directing internal DNS queries to a cloud-based filtering service.
 - *Justification:* Easy to deploy, immediate impact on common web-based threats, minimal cost.

4. **Target: Lack of Visibility and Monitoring.**

- **Attacker Goal:** Operate undetected within the network, conducting reconnaissance or exfiltrating data.
- **Countermeasures (Risk-Based & Practical):**
 - **Enable Syslog on all devices:** Configure all network devices (Core Switch, Access Switches, ASA, Router0) to send their logs to a central syslog server (even a basic Linux server). This provides a centralized repository for events, making it easier to correlate and investigate incidents.
 - *Justification:* Low cost, essential for incident detection and forensics.

- **Basic NetFlow/IPFIX (if supported by devices):** If the Core Switch or ASA supports NetFlow/IPFIX, enable it to collect flow data. This provides valuable insights into network traffic patterns, helping to identify unusual activity.

- *Justification:* Provides visibility into "who is talking to whom," useful for anomaly detection.

3.5 Improved Control Placement

Based on the analysis, here are recommendations for improved control placement:

- **Core Switch:**
 - **ACLs on SVIs:** Place comprehensive ACLs on each VLAN SVI to strictly control interVLAN communication, enforcing a "least privilege" principle. This is the most critical placement for segmenting the network effectively *beyond* basic VLANs.
- **Access Switches:**
 - **Port Security:** Implement port security on all user-facing access ports (FastEthernet0/x) to prevent unauthorized devices from connecting.
 - **DHCP Snooping:** Configure DHCP snooping on access switch ports to prevent rogue DHCP servers and enforce DHCP integrity.
 - **ARP Inspection:** Implement Dynamic ARP Inspection (DAI) on access switches to prevent ARP spoofing attacks.
- **ASA Firewall:**
 - **Granular Outbound ACLs:** Implement more specific outbound ACLs on the ASA's inside interface to control which types of outbound traffic are permitted, rather than relying solely on security levels. This reduces the attack surface for exfiltration.
- **Centralized Logging Server:** Deploy a dedicated server (even a VM) to act as a syslog collector for all network devices. This centralizes logs, improving visibility and incident response capabilities.
- **Potential for Future IDS/IPS:** Consider placing an IDS/IPS sensor either inline behind the ASA (for perimeter monitoring) or mirroring traffic from critical VLANs on the Core Switch to an out-of-band sensor (for internal threat detection).

4. Conclusion

The current college network provides basic connectivity and perimeter security through VLAN segmentation and an ASA firewall. However, the assessment reveals significant weaknesses related to insufficient granular access controls between VLANs, lack of deep threat detection, and limited network access control. By implementing the recommended risk-based countermeasures, particularly strong ACLs on the Core Switch SVIs and port security on access switches, the college can significantly enhance its cybersecurity posture within budget and staff limitations, making the network more resilient against internal and external threats. These recommendations prioritize addressing the most immediate attack vectors and laying a foundation for future security enhancements.



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CYBERSECURITY VIRTUAL INTERNSHIP PROGRAM 2025

REPORT OF PROBLEM STATEMENT FOR CYBER SECURITY INDUSTRY PROJECT

SECURE HYBRID ACCESS MODEL

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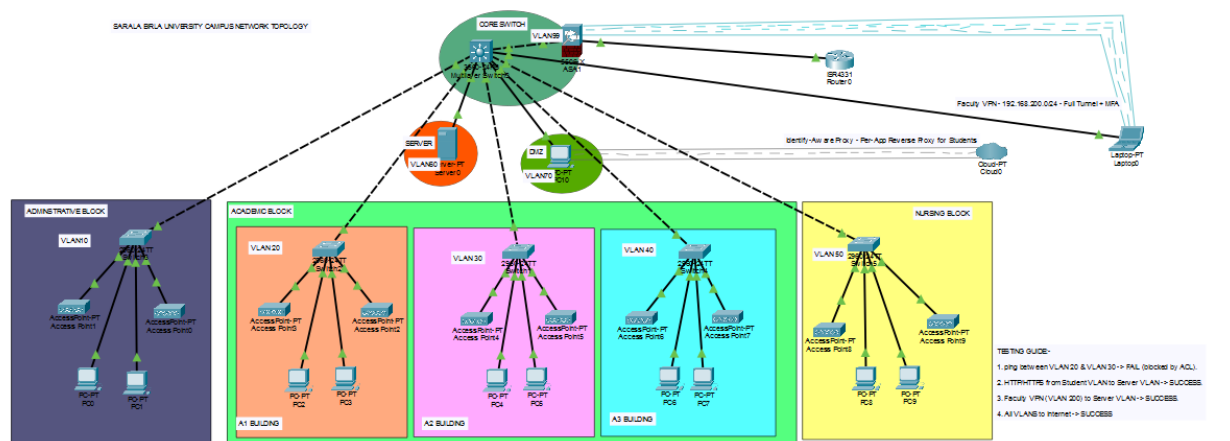
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Cybersecurity Industry Project - Part 2: Secure Hybrid Access Model

1. Introduction and Project Goal

Following the security assessment in Part 1, this report documents the design and implementation of a secure hybrid network model for the college campus. The primary objective is to enable secure and flexible access for faculty and students to internal services, while ensuring these services are **never directly exposed to the internet**. The design accommodates faculty working remotely from home and students using personal devices on or off campus, while balancing simplicity, security, and scalability.

2. Updated Network Topology Diagram



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The following diagram provides a visual representation of the new hybrid architecture.

Key Additions to the Topology:

- **VLAN 60 (SERVERS):** A new VLAN with the subnet **192.168.60.0/24** has been created to house internal servers and services, such as research repositories.
- **VLAN 70 (DMZ):** A new DMZ VLAN with the subnet **192.168.70.0/24** has been added to place components that interact with the internet, such as the Identity-Aware Proxy connector.
- **Faculty Remote Access (VPN):** A "Remote Faculty Laptop" is represented with a dotted cyan line leading to the ASA, symbolizing a **Full-Tunnel VPN** connection from the subnet

192.168.200.0/24. The Core Switch has been configured with a new **VLAN 200** to simulate this VPN segment.

- **Student Remote Access (IAP):** An "Identity-Aware Proxy (IAP) Cloud" is shown with a dotted gray line connected to a DMZ connector, representing a **Per-App Reverse Proxy** model for students.

3. Technical Documentation

3.1 Network Segmentation Based on User Roles

The network design extends the use of VLANs to segment access based on user roles and function.

- **Faculty Network (VLAN 200):** A new VLAN is dedicated to faculty remote access, logically separating their traffic from the main campus network.
- **Student Networks (VLANs 10-50):** The existing VLANs for each building continue to segment student devices. These VLANs are explicitly blocked from accessing other internal VLANs, enhancing security and preventing lateral movement.
- **Internal Services (VLAN 60):** A dedicated VLAN for servers and internal applications isolates them from end-user VLANs, ensuring that only authorized traffic can reach sensitive resources.
- **DMZ (VLAN 70):** The DMZ serves as a buffer zone for services that require controlled interaction with the external network, such as the Identity-Aware Proxy.

3.2 Chosen Secure Access Tools

The design leverages two distinct secure access tools to cater to the unique requirements of faculty and students.

- **VPN (Virtual Private Network):** The chosen solution for faculty remote access. A fulltunnel VPN is used to provide encrypted, secure, and uninterrupted access to the entire campus network, as if the user were physically on-site.
- **Identity-Aware Proxy (IAP):** The chosen solution for student access. This acts as a reverse proxy, providing a secure, per-application access model without granting students network-level access. This approach enforces a **Zero Trust** model, ensuring access is based on identity and context rather than network location.

3.3 Trust Models and Authentication Flows

- **Trust Model:** The design operates on a **Zero Trust** principle, particularly for student access, where no device is trusted by default. All access requests are authenticated and authorized on a per-session or per-application basis, eliminating implicit trust based on network location.
- **Authentication Flows:**
 - **Faculty (VPN):** Faculty members connect using a VPN client. Their access is authenticated using credentials, preferably with **Multi-Factor Authentication (MFA)**, before a secure tunnel is established. Once authenticated, they are assigned

an IP from the 192.168.200.0/24 subnet and can securely access authorized resources.

- **Students (IAP):** Students access applications via a public-facing URL. The IAP intercepts the request and handles the user authentication. Upon successful authentication, the IAP creates a secure, application-specific connection to the internal service, all while ensuring the internal network's IP addresses remain hidden from the student's device.

3.4 Risks, Use Cases, and Fallback Strategies

- **Architectural Justification (Risks and Advantages):**
 - **Risk:** The VPN model for faculty, while convenient, presents a security risk. A compromised faculty device with full network access could allow an attacker to move laterally within the network.
 - **Advantage:** The IAP model for students is highly secure. It minimizes the attack surface by only exposing specific applications, not the entire network. This protects the college's internal infrastructure from the potential security risks associated with BYOD.
- **Use Cases:**
 - **Faculty:** A faculty member at home needs to access a private research repository on the server in VLAN 60. They establish a VPN connection, their laptop gets an IP from the 192.168.200.0/24 range, and an ACL on the Core Switch's Vlan200 interface permits their traffic to reach the server.
 - **Student:** A student wants to check their grades on a personal laptop. They access a web-based portal via the IAP. The IAP verifies their student credentials and grants them access to only the grades application, preventing them from accessing any other network resources.
- **Fallback Strategies:**
 - **VPN Failure:** In the event of a VPN service outage, a contingency plan could involve a simpler web-based portal with read-only access to critical data, mitigating disruption.
 - **IAP Failure:** The IAP is a critical component for student access. If it fails, a fallback could be a simple, static web page explaining the service outage and providing an estimated time to resolution.



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CYBER SECURITY VIRTUAL INTERNSHIP PROGRAM 2025

REPORT OF PROBLEM STATEMENT FOR CYBERSECURITY INDUSTRY PROJECT

SMART WEB ACCESS RESTRICTION

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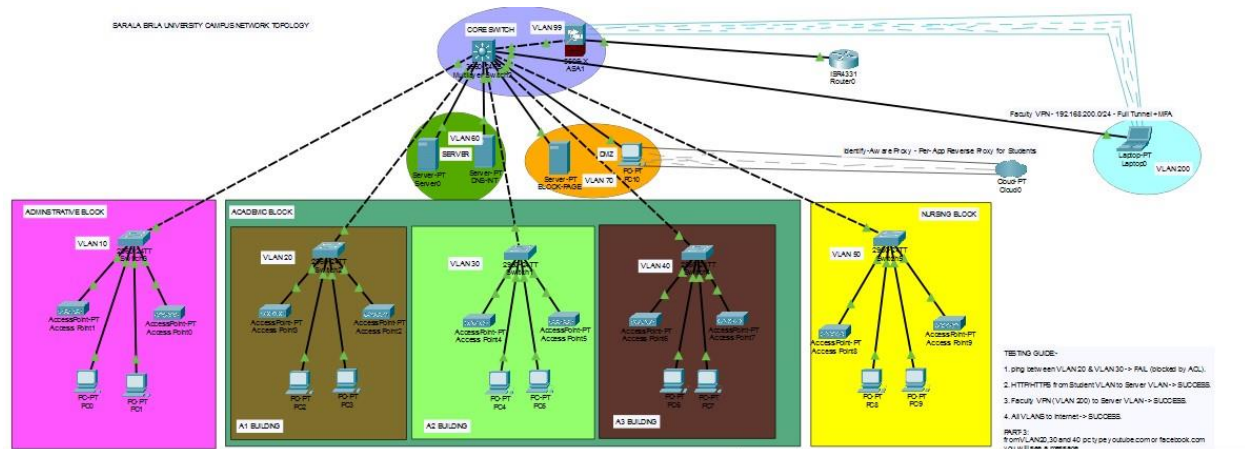
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Cybersecurity Industry Project - Part 3: Smart Web Access Restriction

1. Introduction and Project Goal

This report documents the design and implementation of a web access policy framework to address issues such as video streaming, torrenting, and the use of proxies on the college network. The framework's objective is to intelligently restrict web access without hindering legitimate academic activities or creating user backlash. The policy considers user identity, access time, and content category. The design includes measures to prevent circumvention, along with plans for logging and reporting violations.

2. Updated Topology with Filtering Components



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The network topology from Part 2 was enhanced with new components to enforce and monitor the web access policies. The updated diagram includes a centralized internal DNS server and a dedicated block-page server.

Key Components Added to the Topology:

- **Internal DNS Server (DNS-INT):** A new server was added to the **VLAN 60 Servers** zone with the IP address 192.168.60.53. This server is the core of the web access policy, as it provides a curated DNS service that filters web traffic.
- **Block Page Server (BLOCK-PAGE):** A server was placed in the **VLAN 70 DMZ** with the IP address 192.168.70.10. This server hosts a custom web page that is displayed to users when they attempt to access a blocked site.

These components work together to provide a seamless and transparent blocking mechanism.

3. Web Access Policy Document

The policy framework utilizes a combination of **DNS filtering** and **Access Control Lists (ACLs)** to provide granular control over web access.

3.1 Policy Intent

The policy aims to:

- **Restrict** student access to non-academic content categories like social media (facebook.com, instagram.com), streaming (youtube.com, netflix.com), and gaming (twitch.tv).
- **Prevent** the use of external DNS servers, proxies, and torrenting applications to bypass restrictions.
- **Maintain** unrestricted web access for faculty and staff.
- **Provide** a clear and user-friendly block page that explains why access was denied.

3.2 Comparison of Filtering Solutions

- **DNS Filtering:** This solution is implemented by configuring the internal DNS server to return a specific IP address (192.168.70.10) for blocked domains. This is a simple, cost-effective method that can be centrally managed. However, it can be easily bypassed if users are not forced to use the internal DNS server.
- **Layer 7 (L7) Firewalls:** While not a dedicated appliance in this Packet Tracer simulation, the STUDENT-IN ACL simulates L7 functionality by blocking specific application-layer protocols and ports (e.g., QUIC, proxies). This provides a more robust, low-cost solution than DNS filtering alone.
- **Proxy-Based Filtering:** This method routes all web traffic through a proxy server, allowing for deep content inspection. While very effective, it can be complex to configure and may require a dedicated server, which could be costly.
- **Client-Side Enforcement:** This involves installing software on each end-user device. It is highly effective but difficult to manage and maintain across a large number of devices. The chosen solution—a **hybrid of DNS filtering and L7-style ACLs**—provides a robust, low-cost, and manageable framework for the campus environment.

3.3 Enforcement Logic and Policy Syntax

The policy is enforced through network controls on the **Core Switch** and the new DNS server.

1. Mandatory DNS Enforcement:

- All DHCP pools on the Core Switch were updated to assign the internal DNS server IP (192.168.60.53) to clients.
- An ACL on the Core Switch (STUDENT-IN) explicitly denies all DNS queries (UDP/TCP port 53) that are **not** destined for the internal DNS server, preventing students from using external DNS servers like 8.8.8.8.

2. Web and Application Filtering:

- **DNS Redirection:** The DNS-INT server's records for blocked domains (e.g., youtube.com) are configured to resolve to the BLOCK-PAGE server's IP address (192.168.70.10).

- **ACLs:** The STUDENT-IN ACL is applied inbound on all student VLAN interfaces (Vlan20, Vlan30, Vlan40, Vlan50). This ACL's rules include:
 - deny udp any any eq 443: This rule blocks the QUIC protocol, a bypass for modern streaming.
 - deny tcp any any eq 8080 and deny tcp any any eq 3128: These rules block common proxy ports.
 - deny tcp/udp any any range 6881 6889: These rules block common peer-to-peer (P2P) ports to prevent torrenting.

3. Time-Based Policies:

- A time-range named CLASS_HOURS can be used to apply more restrictive rules during specified periods (e.g., weekdays from 9:00 to 17:00). This allows for flexibility while ensuring network integrity during critical times.

4. Advantages of the Policy Framework

The implemented policy framework offers several key advantages:

- **Effective Anti-Circumvention:** The multi-layered approach of blocking external DNS, common proxy ports, and specific streaming protocols makes it difficult for students to bypass the policy.
- **Role-Based Control:** The policy is applied only to student VLANs, ensuring that faculty and staff retain unrestricted access for their professional needs.
- **Scalability:** The framework is easy to scale; adding a new blocked domain only requires an update to the DNS server's configuration.
- **User-Friendly Experience:** Instead of a simple connection timeout, users are redirected to a custom block page that provides a clear and professional message, reducing confusion and support calls.